HATANGIMANA SIMEON

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PERSONAL PROFILE

Electrical and computer engineer completing an M.S. at Carnegie Mellon University Africa, with a B.Sc. in Electrical Power Engineering and industry experience in transmission, distribution, and MEP design. Focuses on applying machine learning to power system operation, battery health estimation, and microgrid design, with emphasis on methods that remain physically feasible under real network constraints. Seeking doctoral research at the intersection of power systems and data-driven optimisation.

RESEARCH INTERESTS

Power Electronics • Energy Storage Systems & Battery Health Estimation • Microgrid Design & Optimization • Machine Learning for Energy Systems • Energy Access in Developing Regions • IoT & Embedded Energy Systems

EDUCATION

Master of Science in Electrical & Computer Engineering **Aug 2025 – Dec 2026**
Carnegie Mellon University Africa *Kigali, Rwanda*

- Cumulative GPA: **3.62**, improving from 3.38 to 3.83 across successive semesters.
- Focus Areas: Data Science, Machine Learning, Energy Modeling, and Power Electronics.
- Relevant Coursework: Applied Stochastic Processes; Data Inference & ML; Energy Systems Modeling; Power Electronics for Utility Systems; Energy Project Development & Finance; ML for Earth Observation; Energy Data Analytics; Deep Learning; Robotics Principles & Practice.

Bachelor of Science in Electrical Power Engineering **Apr 2021 – Jun 2025**
University of Rwanda, College of Science and Technology *Kigali, Rwanda*

- Cumulative GPA: 3.656 / 4.0.
- Class Rank: 2 of 41.
- Final Year Project: *Sensor-Based Smart Irrigation for Sustainable Farming in Rwanda.*

RESEARCH EXPERIENCE

Battery End-of-Life Research, Graduate Research Project **Jan 2026 – Present**
Carnegie Mellon University Africa • Supervisors: Prof. Sarath Tennakoon and Prof. Barry Rawn *Kigali, Rwanda*

- Building a material flow analysis model from first principles to map lithium-ion battery flows through Rwanda's economy, from import and use through collection to end-of-life processing.
- Evaluating currently available recycling routes, including metallurgical processing, direct recycling, and combined approaches, against the resource and infrastructure constraints specific to Rwanda.
- Collecting and analysing battery performance data using laboratory instrumentation to characterise degradation mechanisms and inform second-life capacity assumptions in the model.

TEACHING EXPERIENCE

Teaching Assistant (Programming for Data Analytics) **Aug 2026 – Dec 2026**
Carnegie Mellon University Africa • Supervisor: Prof. George Okeyo *Kigali, Rwanda*

- Prepared assignments, quizzes, and exams; coordinated with Prof. Okeyo for review and approval.
- Graded assessments, organized students, supervised examinations, and held office hours.
- Supported lecture delivery and in-class activities as needed.

Teaching Assistant (Graduate Bridge Program) **Feb 2026 – May 2026**
Carnegie Mellon University Africa • Supervisor: Prof. Emmanuel Ndayishimiye *Kigali, Rwanda*

- Supported two cohorts in foundational engineering and programming, including a combined cross-campus cohort exceeding 100 participants.

RESEARCH PROJECTS

Battery Capacity Passport **Aug 2025 – Dec 2025**
Collaborative Project • Supervisor: Prof. Barry Rawn

- Built a software-first, hardware-enabled battery certification concept for Rwanda's e-mobility ecosystem.
- Proposed tamper-proof digital "passports" (QR + cryptographic signature) from garage diagnostics + lab/cloud verification, reporting SoH, usable capacity, and internal resistance.
- Produced market and financial analysis showing feasibility under $200k CAPEX (IRR 16%, NPV $127,435; payback year 7) and alignment with Regulation N°002/2018 on e-waste.

MicroGrid Modeller **Jan 2026 – May 2026**
Supervisor: Prof. Nathan Williams

- Built a Streamlit application that takes operator-supplied system parameters and returns techno-economic analysis of a proposed microgrid, including dispatch simulation, interactive dashboards, and downloadable results.
- Designed to make techno-economic assessment accessible to operators without specialist modelling training, addressing a practical barrier to microgrid deployment in energy-limited settings.

Energy Outlook Navigator **Jan 2025 – Mar 2026**
Independent Project

- Built a Streamlit application on United Nations energy datasets that lets users visualise any country's current energy profile and adjust input assumptions to explore forward scenarios.

PROFESSIONAL EXPERIENCE

Electrical Design Professional Intern **Nov 2024 – Jul 2025**
QuestAfrica Ltd *Nyarutarama, Kigali*

- Supported integrated MEP electrical designs and engineering calculations to improve building energy efficiency.
- Prepared technical reports and provided site support to align electrical designs with field implementation.

Power Transmission and Distribution Intern **Jan 2023 – Apr 2023**
Rwanda Energy Group (REG / EUCL) *Rubavu, Rwanda*

- Analyzed transmission and distribution performance data across the Western Province network.
- Supported infrastructure maintenance and prepared statistical reports for technical assessment and planning.

Engineering Intern — Kabuye Sugar Works, Kigali Jun 2024 – Jul 2024
Electrical installations, operation control, and maintenance of motors and generators in sugar production.

Engineering Intern — Gasabo 3D, Kigali Jan 2024 – May 2024
Electrical consultation on the MPAZI rehousing project: lighting plans, single-line diagrams, load calculations, and lightning protection designs (AutoCAD, DIALux evo).

LEADERSHIP & SERVICE

President **Dec 2025 – Present**
Energy and Embedded Systems Club — CMU Africa *Kigali, Rwanda*

- Lead club activities, coordinate technical workshops, and support member engagement in energy systems, electronics, and embedded technologies.
- Serve as student organiser for the 2026 Open Source in Energy Access Symposium, hosted at CMU Africa with EnAccess and the World Resources Institute, recruiting and coordinating 10 student volunteers and preparing approximately 50 student participants for an event drawing over 200 attendees.

General Coordinator (Volunteer) **Jan 2021 – Jan 2023**
EmpowerForum CST *Kigali, Rwanda*

- Coordinated knowledge-sharing and student engagement activities for a community focused on electrical power engineering.

AWARDS & HONOURS

Government of Rwanda Scholarship (M.S., Carnegie Mellon University Africa) — 2025–2026
Government of Rwanda Scholarship (B.Sc., University of Rwanda) — 2021–2025
Community Spirit Award — Carnegie Mellon University Africa, 2025
Credible Leadership Applications (CCL-3) — University of Rwanda, 2025 (recognition of two years as General Coordinator of EmpowerForum)

TECHNICAL SKILLS

Programming Languages	Python, MATLAB, C
Machine Learning & Data	TensorFlow, PyTorch, scikit-learn, NumPy, Pandas, Seaborn, Matplotlib
Engineering Software	AutoCAD, DIALux, PowerWorld, Microsoft Excel, Word, PowerPoint
Project Management	Gantt chart planning, cross-functional collaboration, technical reporting, public speaking
Hardware & Systems	Sensor-based systems, embedded IoT devices, MEP electrical design

CERTIFICATIONS & AFFILIATIONS

Duolingo English Test (English Proficiency Certificate) — 2026 (valid to 2028)
IEEE Power & Energy Society, Student Member — 2026
Battery Box Handling — Collective Energy Africa & BYD, 2023
Graduate Bridge Program — Carnegie Mellon University Africa, 2024 (six-week programme in IoT and embedded systems)
GeoPandas Certification — Modern GIS, 2026

LANGUAGES

English	**French**	**Kinyarwanda**	**Swahili**
Fluent	Elementary	Native	Native